Electrolyser



Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Electrolyser

What it is used for

An electrolyser uses electrical energy to split water or another approved feedstock into chemical products, such as hydrogen and oxygen in water electrolysis. Correct fluid preparation, electrical control, gas handling, and ventilation help maintain process efficiency and reduce fire, explosion, chemical, and electrical hazards.

Main parts

- Electrolyser stack or cell assembly
- Power supply and current or voltage controls
- Feedwater or electrolyte reservoir and circulation system
- Gas outlets, separators, and collection lines
- Pressure, temperature, and flow sensors
- Ventilation, cooling system, protective enclosure, and emergency shutdown

Before you start

1. Operate the electrolyser only if trained and authorised for the specific technology and process.
2. Read the equipment manual, process diagram, safety data sheets, and site emergency procedures.
3. Confirm the feedwater or electrolyte meets the manufacturer's specified purity, concentration, and compatibility requirements.
4. Inspect the stack, hoses, fittings, valves, seals, electrical cables, sensors, and protective enclosure for leaks or damage.
5. Verify that ventilation, cooling, pressure relief, gas detection, and emergency shutdown systems are available and functional.
6. Check that hydrogen and oxygen outlets are correctly identified and routed to approved, separate handling systems as designed.
7. Ensure the work area is free of ignition sources and that required PPE and gas monitoring are available.

How to use

1. Confirm that the system is correctly assembled, all required guards are fitted, and access to emergency shutdown controls is clear.
2. Fill the feedwater or electrolyte reservoir using the approved material and specified procedure.
3. Open required valves and establish the prescribed circulation, cooling, and ventilation conditions before energising the stack.

4. Check that gas outlets, pressure relief paths, and collection or vent lines are open and correctly connected.
5. Switch on the control system and verify sensor readings, interlocks, alarms, and operating status.
6. Set the approved current, voltage, temperature, pressure, and flow limits for the equipment and process.
7. Energise the electrolyser using the manufacturer's startup sequence and increase load only as specified.
8. Monitor voltage, current, temperature, pressure, fluid level, flow, gas output, and alarms at the required intervals.
9. Check for leaks, unusual vibration, unexpected heating, abnormal gas flow, or signs of cross-contamination between gas streams.
10. If a leak, gas alarm, cooling failure, excessive pressure, or abnormal temperature occurs, initiate the approved shutdown and emergency response.
11. Do not tighten fittings or repair leaks while the system is energised or pressurised.
12. At the end of the run, reduce load and shut down according to the manufacturer's sequence; isolate electrical power when required.
13. Allow the system to reach its specified safe state, then isolate feed and gas lines according to the approved procedure.
14. Record operating parameters, alarms, maintenance observations, and any deviations from the approved process.

⚠ Safety

- Hydrogen is highly flammable and can form explosive mixtures with air; keep ignition sources away and maintain approved ventilation.
- Do not mix hydrogen and oxygen streams or modify gas piping without authorised engineering approval.
- Never block gas outlets or pressure-relief devices, and do not exceed rated pressure or temperature limits.
- Use only compatible materials and the specified electrolyte; corrosive electrolytes require appropriate chemical PPE and handling procedures.
- Treat the stack, terminals, and power connections as hazardous electrical equipment during operation and maintenance.
- Use approved leak-detection methods and follow site procedures for gas alarms, ventilation failure, or suspected leaks.
- Depressurise and isolate the system according to the manufacturer's procedure before maintenance; never open a pressurised cell or line.
- Follow applicable hazardous-area, gas-storage, fire-safety, and emergency-response requirements.

Care & storage

- Inspect stack seals, fittings, hoses, valves, and electrical connections at the prescribed intervals.

- Monitor feedwater or electrolyte quality and replace or replenish it according to the manufacturer's instructions.
- Check cooling, circulation, ventilation, pressure relief, and gas detection systems regularly.
- Keep the equipment exterior clean and dry without allowing cleaning fluids to enter electrical or gas components.
- Maintain operating logs and arrange scheduled servicing, leak checks, and sensor calibration.

Common mistakes to avoid

- Using feedwater or electrolyte outside the specified quality or concentration limits.
- Starting the stack before establishing circulation, cooling, ventilation, and safe gas routing.
- Ignoring pressure, temperature, gas alarms, or suspected leaks.
- Opening or servicing pressurised or energised equipment without approved isolation procedures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.